

Roll No. 2200186023

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**B. Pharma. 7th Semester
(Regular/Re-Appear/Mercy Chance)
Examination – December, 2025**

INDUSTRIAL PHARMACY II

Paper : BP702T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks. Attempt any *seven* questions out of nine from Section-B, each carry 5 marks and *two* questions out of three from Section-C, each carry 10 marks.

SECTION – A

Note : Attempt *all* questions.

$10 \times 2 = 20$

1. (a) Write the introduction about platform technology.
- (b) What do you mean by investigator's brochure ?

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- (c) Write a note on quality management.
- (d) What do you mean quality control ?
- (e) Write a note on quality by design (QbD).
- (f) What is pharmacology ?
- (g) Explain CDSCO.
- (h) Write a note BCIL.
- (i) Discuss about clinical research.
- (j) Write a note on confidentiality agreement.

SECTION – B

Note : Attempt any *seven* questions.

7 × 5 = 35

- ✓ 2. Discuss about certificate of pharmaceutical product.
- 3. Write a note on state licensing authority.
- ✗ 4. What are the responsibility of regulatory affairs department ?
- ✗ 5. Discuss about FDA submission for approval of drug.
- ✓ 6. Explain approval procedures of new drugs.
- ✓ 7. Discuss about total quality management.

- ✓ 8. Explain general considerations of oral products in pilot plant scale up.
- ✓ 9. Explain general considerations of Investigational New Drug (IND).
- ✓ 10. Write a note on analytical method transfer.

SECTION – C

Note : Attempt any *two* questions.

$2 \times 10 = 20$

- 11. Discuss about introduction, historical overview and authorities of regulatory affairs.
 - 12. Discuss about general considerations of pilot plant scale up techniques.
 - 13. Discuss WHO guidelines for technology transfer.
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Department of Pharmaceutical Sciences, I.G.U., Meerpur, Rewari
B. Pharm 7th semester, Industrial Pharmacy-II (BP 702T)
Ist sessional Exam, September 2025

Max Marks: 30

Time: 1 hr

Attempt all questions from part A (10x1)

Attempt 2 questions from part B (2x5)

Attempt 1 question from part C (1x10)

Part-A

- (i). SUPAC (ii) What scale up parameters to be considered for capsules. (iii) Clinical Research Protocols (iv) Cross Latin square cross over design (v) Write down name of current Drug Controller General of India and his duties.

Part B

- (i). Write about Scale up and Post Approval Changes.
(ii). Elaborate the historical milestones of Indian Pharmaceutical Regulatory Affairs.
(iii). Explain importance and format of Investigation New Drug Application.

Part-C

- (i). Describe in detail functions and organizational structure of CDSCO.
(ii). Discuss in detail about pilot scale up consideration for tablets.

Department of Pharmaceutical Sciences, I.G.U., Meerpur, Rewari
B. Pharm 7th semester, Industrial Pharmacy-II (BP 702T)
2nd sessional Exam, November 2025 (Exam Date 25.11.2025)

Time: 1 hr

Max Marks: 30

Attempt all questions from part A (5x2), Attempt 2 questions from part B (2x5), Attempt 1 question from part C (1x10)

Part A

(i). Describe "4M" affecting quality of a pharmaceutical product. (ii) DMAIC (iii) CoPP (iv) Benefits of NABL Accreditation (v) Critical Quality Attributes.

Part B

- (i). Briefly describe WHO guidelines of Technology transfer.
- (ii) What are the major responsibilities of state licensing authority?
- (iii) Write a note on ISO 9000 series of Quality Management Standards.

Part C

- (i). Describe in detail regulatory requirements and approval procedure for new drugs.
- (ii) Discuss importance and different aspects of Quality by Design used in pharmaceutical industry.

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**B. Pharma. 7th Semester
(Regular/Re-Appear/Mercy Chance)
Examination – December, 2025**

INSTRUMENTAL METHODS OF ANALYSIS

Paper : BP701T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks, Attempt any *two* questions out of three from Section-B, each carry 10 marks and *seven* questions out of *nine* from Section-C, each carry 5 marks.

SECTION – A

1. Attempt *all* questions : $2 \times 10 = 20$

- (a) Give *two* applications of Electrophoresis.
- (b) Give the principle of Atomic Absorption Spectroscopy.

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- (c) Give differences between TLC and Paper Chromatography.
- (d) Define Fermi Resonance in IR spectroscopy.
- (e) Define Singlet and Triplet excited states.
- (f) Give principle of Gel Chromatography.
- (g) Define Isocratic and Gradient elution in HPLC.
- (h) Give principle of UV-Vis spectroscopy.
- (i) Give *two* limitations of Flame photometry.
- (j) Define $E^{1\%}_{1\text{cm}}$ value of a compound.

SECTION – B

Note : Answer any *two* out of the following three questions :

- 2. Discuss the principle, instrumentation and applications of double beam UV - Vis spectrophotometer. Draw a neat and well labelled flow diagram of it. 10
- 3. Discuss the following :
 - (a) Factors affecting vibrational frequency in IR spectroscopy. 7
 - (b) Theory and *two* applications of affinity chromatography. 3

4. Discuss the principle, instrumentation and applications of Gas Chromatograph. Draw a neat and well labelled flow diagram of it. 10

SECTION - C

Note : Answer any *seven* out of the following nine questions :

5. Define Beer-Lambert's law. Discuss various causes of deviation from it. 5
6. Discuss principle and methodology of Nepheloturbidometry. 5
7. Write a note on any *three* detectors of HPLC unit. 5
8. Discuss principle and methodology of Thin Layer Chromatography. 5
9. Define Electrophoretic mobility. Discuss the factors affecting it. 5
10. Write a note on the burners used in flame photometry. 5
11. Discuss principle and methodology of ion exchange chromatography. 5
12. Write a note on any *three* detectors of IR spectrophotometer. 5
13. Discuss any *two* interferences observed in AAS. How they are corrected ? 5

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari
B. Pharm. 7th Semester, 1st Sessional Examination, Sept, 2025
Instrumental Methods of Analysis (Theory) (Paper Code- BP701T)

Date: Sept 29, 2025

Duration: 1 hr

Total marks: 30

Question no.1 is compulsory. Draw diagram wherever required.

1. Define the following.

(5×2=10 marks)

- Write the advantages and disadvantages of column chromatography.
- What is matrix interference? Give examples.
- What do you mean by fluorescence?
- Define auxochrome and chromophore giving examples.
- Give the pharmaceutical application of IR Spectroscopy.

- TLC
- Explain the effect of solvent in IR
- Indication

Attempt any two questions.

(2×5=10 marks)

- Write the principle, sources of radiation and detectors used in UV spectroscopy.
- Discuss the types of paper chromatography with its applications.
- What are singlet and triplet electronic states? Describe various factors affecting fluorescence.

Attempt any one question.

(1×10=10 marks)

- Write a note on a) Detectors used in IR b) Sample handling in IR Spectroscopy
- Describe the methodology and applications of TLC.

Department of Pharmaceutical Sciences, IGU Meerpur,

B. Pharm. 7th Sem, 2nd Sessional, Nov, 2025

Instrumental Methods of Analysis (Paper Code- BP701T)

Date: Nov 24, 2025 Duration: 1 hr

Total marks: 30

Question no.1 is compulsory. Draw diagram wherever required.

1. Define the following. (5×2=10 marks)

- a. Write two applications of gel permeation chromatography.
- b. Write two examples each of cation and anion exchangers
- c. State the advantages and disadvantages of agarose and polyacrylamide gels.
- d. Write about total combustion burner with diagram.
- e. Explain the principle of flame photometry.

Attempt any two questions. (2×5=10 marks)

5. Explain in detail various factors affecting ion exchange.
6. Explain various types of interferences in AAS.
7. Explain the theoretical principle and applications of affinity chromatography.

Attempt anyone questions. (1×10=10 marks)

5. Define derivatization. Write a note on different derivatization techniques in gas chromatography and its significance.
6. Explain various types of detectors used in HPLC. Also explain the factors affecting resolution in HPLC.

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**B. Pharma. 7th Semester
(Regular/Re-Appear/Mercy Chance)
Examination – December, 2025**

NOVEL DRUG DELIVERY SYSTEM

Paper : BP704T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks, Attempt any *seven* questions out of nine from Section-B, each carry 5 marks and *two* questions out of three from Section-C, each carry 10 marks.

SECTION – A

1. Attempt *all* questions.

10 × 2 = 20

(a) Write a note on permeation enhancer.

(b) Explain the advantage of natural polymers.

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- (c) What is microencapsulation ?
- (d) Write a note on microspheres.
- (e) What do you mean by NDDS ?
- (f) Write the merits and demerits of mucosal drug delivery system.
- (g) Discuss about osmotic pump.
- (h) Write a note on diffusion.
- (i) Write a note on metered dose.
- (j) Discuss biological properties of drug relevant to CDDS.

SECTION – B

Note : Answer any *seven* out of the following nine questions :

$$7 \times 5 = 35$$

- 2. What is the factor affecting permeation in transdermal drug delivery system ?
- 3. Discuss in detail about nebulizer.
- 4. Write a note on intraocular barriers.
- 5. Explain in detail about liposomes.
- 6. Discuss the method of microencapsulation and its application.

- ✓ 7. Discuss about implantable drug delivery system.
- ✓ 8. Explain the classification and properties of polymers.
- ✓ 9. Write a note on control drug delivery system.
10. Discuss the principle of bioadhesion and mucoadhesion.

SECTION – C

Note : Answer any *two* out of the following three questions : $2 \times 10 = 20$

11. Discuss in details about targeted drug delivery system.
 12. Explain gastroretentive drug delivery system in detail.
 13. What are Intrauterine drug delivery system discuss in detail ?
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DEPARTMENT OF PHARMACEUTICAL SCIENCES, INDIRA GANDHI UNIVERSITY, MEERPUR, REWARI
First Sessional Theory Examination of B. Pharmacy 7th Semester –October, 2025
Subject: Novel drug delivery system Theory (Paper Code: BP704T)

Time: 1.0 hr

Max. Marks =30

Attempt all questions

Q1	Give the answers of all questions				5 × 2 = 10
	i.	Differentiate matrix and reservoir	iv	What is the range of microcapsules? What size of particles are suitable for air suspension coating?	
	ii.	State applications of polymers in CRDDS	v	Write 2 advantages and 2 disadvantages of mucoadhesive DDS.	
	iii.	Name the three immiscible phases in coacervation.			
Q2.	Attempt any two question (Short answer type questions)				2 × 5 = 10
	a)	Define polymer and classify them.			
	b)	Describe any 2 theories of mucoadhesion.			
	c)	Explain osmotically pressure activated DDS.			
Q3	Attempt any one question (Long answer type questions)				1 × 10 = 10
	a)	Explain the design involved in controlled release formulations.			
	b)	Describe in detail buccal and sublingual drug delivery with their advantages and disadvantages.			

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**B. Pharma. 7th Semester
(Regular/Re-Appear/Mercy Chance)
Examination – December, 2025**

PHARMACY PRACTICE

Paper : BP703T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks.

Attempt any *two* questions out of three from Section-B, each carry 10 marks and *seven* questions out of nine from Section-C, each carry 5 marks.

SECTION – A

Note : Attempt *all* questions.

$10 \times 2 = 20$

1. (a) Enlist types of hospitals on clinical basis.
(b) Write any *two* responsibilities of a pharmacist in a hospital.

P. T. O.

- (c) What are controlled drugs ? Give *one* example.
- (d) Name any *two* drugs that requires therapeutic drug monitoring.
- (e) Mention classes of drugs that are most critically monitored by automatic stop orders in hospitals.
- (f) What are open-ended questions in patient counselling ? Give an example.
- (g) How do people use and misuse OTC medicines in India ?
- (h) How does protein binding of a drug influence the pharmacokinetics ?
- (i) Why specific gravity of urine is measured in Urinalysis ?
- (j) What is the clinical significance of Erythrocyte Sedimentation Rate (ESR) blood test ?

SECTION – B

Note : Attempt any *two* questions.

$2 \times 10 = 20$

- ✓ 2. Explain in detail about the legal requirements for establishment and maintenance of a drug store.
- ✓ 3. Discuss in detail about monitoring of patient's medication adherence.

4. Classify Investigation new drugs. How a hospital pharmacist plays a key role in investigational use of drugs.

SECTION – C

Note : Attempt any *seven* questions.

7 × 5 = 35

5. Discuss how enzyme inducers and enzyme inhibitors influence pharmacokinetic interactions ?
6. Differentiate between spontaneous case reports and record linkage studies in detecting ADRs.
7. What are satellite pharmacy services ? Describe the role of pharmacist in bed side pharmacy.
8. Differentiate between a hospital formulary and a hospital drug list.
9. Explain the role of a community pharmacist in promoting medication adherence.
10. Describe code of ethics for community pharmacist.
11. Discuss in brief about the importance of effective communication between pharmacists and prescribers in patient care.

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12. What are income accounts ? Describe briefly about implementation of budget.
 13. State the objectives of inventory management.
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Subject: Pharmacy Practice Time: 1.5 Hrs B.Pharmacy 7th sem, 1st Sessional Max Marks: 30
Note: Section A is compulsory. Attempt any two questions from section B and one question from section C.

Section A (Very Short Answer Questions)

(2x5=10)

Q.1)

- i) What is idiosyncrasy?
- ii) Differentiate between retail and wholesale drug stores.
- iii) Define hospital formulary?
- iv) What is TDM?
- v) What is medication non-adherence?

Section B (Short Answer Questions)

- Q.2 Write a note on adverse drug reaction. (05 Marks)
- Q.3 Write down classification of hospitals. (05 Marks)
- Q.4 What is need for the patient medication history interview? Briefly explain with form. (05 Marks)

Section C (Long Answer Questions)

- Q.5 Define hospital. Write its organization structure. Explain services of hospital in details. (10 Marks)
- Q.6 Write a note on drug distribution system in a hospital. (10 Marks)

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari
B. Pharm. 7th Semester, 2nd Sessional Examination, 26th November, 2025
Paper Code – BP703T; Pharmacy Practice (Theory)

Duration: 1 hr

Total marks: 30
(10 marks)

1. Give a brief description of the following:

- a. Automatic stop order
- b. Patient counselling
- c. Code of ethics for community pharmacy
- d. PRN and STAT
- e. Clinical review

2. Attempt any one out of two.

(10 marks)

- a. Discuss in brief about budget preparation and implementation.
- b. Elaborate on the organisation and functions of the pharmacy and therapeutic committee.

3. Write a note on the following (any two):

(5+5 marks)

- a. Discuss different healthcare software used in pharmacy.
- b. Define OTC drugs with special emphasis on the rational use of common OTC drugs.
- c. Define therapeutic drug monitoring. Write the note on the role of pharmacist intervention in TDM.